

LINEAR INDEPENDENCE OF FOUR TIME–FREQUENCY TRANSLATES AT A RESONANT NON-LATTICE CONFIGURATION

PRINCIPIA MATH

ABSTRACT. Let Z denote the Zak transform, a an irrational number and j an integer. We study the four-point configuration $\Lambda_{a,j} = \{(0, 0), (1, 0), (0, 1), (a, a + j)\}$, which is *resonant*: the translation of $\mathbb{T}^2$ that governs it is not ergodic, and $\mathbb{T}^2$ fibres into invariant circles. We prove two statements. First, unconditionally: if $g \neq 0$ has a continuous Zak transform — in particular if g is a nonzero Schwartz function — then the translates indexed by $\Lambda_{a,j}$ are linearly independent for every irrational a and every $j \notin \{-1, 1\}$. Taking $a = \sqrt{2}$, $j = 0$ settles the HRT subconjecture of Heil and Speegle [8, Conjecture 2] exactly as stated. Second, conditionally on the theorem of Iwanik–Lemańczyk–Rudolph on absolutely continuous cocycles over irrational rotations, the same conclusion holds for *every* nonzero $g \in L^2(\mathbb{R})$, with no continuity or decay hypothesis.

The two statements are proved by two different routes that share a reduction and a common central identity, the root count $N(\theta) = j + 1$. The unconditional route obtains it from the invariance of the degree of a circle map under precomposition with a rotation; the L^2 route, where no degree is available, obtains it from a spectral obstruction. We also show that $j = \pm 1$ is precisely where both routes saturate. The L^2 route is formally verified in Lean 4 against Mathlib; see Section 8.

1. INTRODUCTION

The HRT conjecture [7] asserts that for every nonzero $f \in L^2(\mathbb{R})$ and every finite set of distinct points $\{(a_k, b_k)\} \subset \mathbb{R}^2$ the functions $e^{2\pi i b_k x} f(x - a_k)$ are linearly independent. It is known for at most three points [7], for subsets of a lattice [11], for configurations lying on two parallel lines [4, 5], for almost every $(1, 3)$ configuration [12], and for various restricted window classes [2, 13]. The smallest configuration resisting all of these is

$$\Lambda_0 = \{(0, 0), (1, 0), (0, 1), (\sqrt{2}, \sqrt{2})\},$$

singled out by Heil and Speegle as the *HRT subconjecture* [8, Conjecture 2]: if $g \in \mathcal{S}(\mathbb{R}) \setminus \{0\}$, then $g(x)$, $g(x - 1)$, $e^{2\pi i x} g(x)$, $e^{2\pi i \sqrt{2} x} g(x - \sqrt{2})$ are linearly independent.

Throughout, for $x, \omega \in \mathbb{R}$ write $(T_x g)(t) = g(t - x)$ and $(M_\omega g)(t) = e^{2\pi i \omega t} g(t)$, and put

$$\Lambda_{a,j} = \{(0, 0), (1, 0), (0, 1), (a, a + j)\}, \quad a \in \mathbb{R} \setminus \mathbb{Q}, \quad j \in \mathbb{Z},$$

so that $\Lambda_0 = \Lambda_{\sqrt{2},0}$. Our results are the following two theorems.

Theorem 1 (Continuous windows; unconditional). *Let a be irrational and $j \in \mathbb{Z} \setminus \{-1, 1\}$. Let $g \in L^2(\mathbb{R})$ be nonzero and suppose the quasi-periodic extension (1) of Zg is continuous on all of $\mathbb{R}^2$. Then*

$$g, \quad T_1g, \quad M_1g, \quad M_{a+j}T_ag$$

are linearly independent. This applies in particular to every nonzero Schwartz g , and more generally to every nonzero g in the Wiener amalgam $W(C, \ell^1)$. Taking $a = \sqrt{2}$ and $j = 0$ proves [8, Conjecture 2].

Theorem 2 (Arbitrary L^2 windows; conditional on Theorem 5). *Let a be irrational and $j \in \mathbb{Z} \setminus \{-1, 1\}$. Assume Theorem 5 below. Then for every nonzero $g \in L^2(\mathbb{R})$ the four functions $g, T_1g, M_1g, M_{a+j}T_ag$ are linearly independent. No continuity or decay hypothesis on g is used.*

Theorem 5 is a published theorem of Iwanik, Lemańczyk and Rudolph, not a conjecture; we nevertheless flag Theorem 2 as conditional because that input is quoted rather than reproved here, and because it is the one step of the present development that is not formally verified (Section 8).

Remark 3 (Λ_0 is not reachable from the solved cases). Affine symplectic maps preserve HRT dependences, and the difference vectors of Λ_0 have symplectic invariants $1, \sqrt{2}, -\sqrt{2}$; membership in a lattice would force these into a common $d\mathbb{Z}$, which is impossible since $1/\sqrt{2}$ is irrational. Parallelism and collinearity are affine invariants and fail for Λ_0 . Hence Linnell’s theorem, the two-parallel-line results and the collinear results cannot be applied after any change of coordinates.

Remark 4 (Scope). Theorems 1 and 2 concern one resonant family of four-point configurations. They are not a proof of the HRT conjecture for arbitrary four-point sets: the hypothesis that the fourth point has the form $(a, a + j)$ with $j \in \mathbb{Z}$ is what produces the invariant-circle decomposition, and is essential to everything below.

The external input for Theorem 2 is the following. If $R_at = t + a$ is an irrational rotation of $\mathbb{T}$ and $\xi : \mathbb{T} \rightarrow \mathbb{T}$, let $V_{\xi,a}f(t) = \xi(t)f(R_at)$ for $f \in L^2(\mathbb{T})$.

Theorem 5 (Iwanik–Lemańczyk–Rudolph [9]). *Suppose a is irrational and $\xi : \mathbb{T} \rightarrow \mathbb{T}$ is absolutely continuous, has nonzero topological degree, and has derivative of bounded variation. Then $V_{\xi,a}$ has Lebesgue maximal spectral type. In particular $V_{\xi,a}$ has no eigenvectors.*

This is the form of the main spectral statement of [9]; see also [10, Theorem 4]. We apply it only to real-analytic circle-valued ξ , for which the regularity hypotheses are immediate.

Organisation. Section 2 performs the Zak reduction in L^2 and isolates the fibre dichotomy. Section 3 proves the central identity $N(\theta) = j + 1$ twice, once under continuity and once under Theorem 5. Section 4 extracts the mean condition from Jensen's formula. Section 5 gives two endgames and proves the theorems. Section 6 explains why $j = \pm 1$ resists both routes. Section 7 compares with the recent work of Guan and Okoudjou. Section 8 describes the Lean 4 formalisation. Section 9 sets out the division of credit with respect to Oussa's program.

2. THE ZAK REDUCTION AND THE INVARIANT FIBRES

Normalise the Zak transform by

$$Zg(t, \omega) = \sum_{n \in \mathbb{Z}} g(t - n) e^{2\pi i n \omega},$$

interpreted on a dense subspace and extended by unitarity from $L^2(\mathbb{R})$ onto $L^2([0, 1)^2)$. Its quasi-periodic extension satisfies, almost everywhere,

$$(1) \quad Zg(t + 1, \omega) = e^{2\pi i \omega} Zg(t, \omega), \quad Zg(t, \omega + 1) = Zg(t, \omega),$$

and

$$(2) \quad Z(M_b T_x g)(t, \omega) = e^{2\pi i b t} Zg(t - x, \omega - b).$$

Assume towards a contradiction that

$$(3) \quad Ag + BT_1 g + CM_1 g + DM_{a+j} T_a g = 0$$

with coefficients not all zero. If one coefficient vanishes, (3) is a dependence among at most three time–frequency translates. For one or two points this is immediate; for three points it is excluded by the theorem of Heil–Ramanathan–Tapiwala [7]. (Equivalently, for three points in general position an element of $SL(2, \mathbb{R})$ carries the difference vectors to $(1, 0)$ and $(0, s)$ with $s \neq 0$, placing the configuration inside the lattice $\mathbb{Z} \times s\mathbb{Z}$, where Linnell's theorem [11] applies; metaplectic operators are unitary on L^2 , preserve the Schwartz class, and carry time–frequency translates to time–frequency translates up to unimodular factors, so dependences correspond.) Hence

$$(4) \quad ABCD \neq 0.$$

Put $G = Zg$. By (1), (2) and $j \in \mathbb{Z}$, (3) becomes

$$(5) \quad p(t, \omega) G(t, \omega) = -D e^{2\pi i(a+j)t} G(t-a, \omega-a), \quad p(t, \omega) = A + B e^{-2\pi i \omega} + C e^{2\pi i t}.$$

Note that j has disappeared from the shift and survives only in the prefactor.

The translation $(t, \omega) \mapsto (t - a, \omega - a)$ of $\mathbb{T}^2$ is *not* ergodic: $e^{2\pi i(\omega - t)}$ is a nonconstant invariant character. Its ergodic components are the circles $\{\omega = t + \theta\}$, on which the translation acts as the rotation by a ; the parameter $\theta = \omega - t \bmod 1$ is well defined on $\mathbb{T}^2$ and each fibre is a single circle. For $\theta \in \mathbb{T}$ set

$$G_\theta(t) = G(t, t + \theta), \quad P_\theta(t) = A + B e^{-2\pi i(t+\theta)} + C e^{2\pi i t}.$$

The coordinate change $(t, \theta) \mapsto (t, t + \theta)$ preserves Haar measure on $\mathbb{T}^2$. Consequently, for almost every θ the restriction of G_θ to $[0, 1)$ lies in $L^2([0, 1))$ and the following hold for almost every t :

$$(6) \quad G_\theta(t + 1) = e^{2\pi i(t + \theta)} G_\theta(t),$$

$$(7) \quad G_\theta(t - a) = -D^{-1} e^{-2\pi i(a + j)t} P_\theta(t) G_\theta(t).$$

Moreover

$$(8) \quad \|g\|_2^2 = \int_{\mathbb{T}} \int_0^1 |G_\theta(t)|^2 dt d\theta.$$

If Zg is continuous, (6) and (7) hold for *every* t and every θ , and G_θ is continuous.

Call a fibre *live* if G_θ is not the zero element of $L^2([0, 1))$. Since $g \neq 0$, (8) shows the set of live fibres has positive measure.

Lemma 6 (Fibre dichotomy). *For almost every θ , either $G_\theta = 0$ almost everywhere, or $G_\theta(t) \neq 0$ for almost every t .*

If in addition Zg is continuous, then for every θ either $G_\theta \equiv 0$ or G_θ is nowhere zero; the set $C_1 = \{\theta : G_\theta \text{ is nowhere zero}\}$ is open, and $C_1 = \emptyset$ forces $g = 0$. In particular, for nonzero g with continuous Zak transform, C_1 contains an interval.

Proof. Taking absolute values in (7),

$$(9) \quad |G_\theta(t - a)| = |D|^{-1} |P_\theta(t)| |G_\theta(t)|.$$

By (6) the function $|G_\theta|$ is 1-periodic. The trigonometric polynomial P_θ is not identically zero and so has finitely many zeros; hence $|P_\theta| > 0$ almost everywhere, and (9) shows the zero set of $|G_\theta|$ is invariant modulo null sets under the irrational rotation $t \mapsto t - a$. Ergodicity gives measure zero or one.

For the continuous statement, (7) shows $G_\theta(t_0) = 0$ forces $G_\theta(t_0 - a) = 0$, hence $G_\theta(t_0 - na) = 0$ for every $n \geq 0$; no division occurs. As a is irrational this orbit is dense modulo 1, and $|G_\theta|$ is 1-periodic, so continuity gives $G_\theta \equiv 0$. By (1) the modulus $|G|$ is a well-defined continuous function on $\mathbb{T}^2$, so $\{|G| > 0\}$ is open, and C_1 is its image under the bundle projection $\mathbb{T}^2 \rightarrow \mathbb{T}$, $(t, \omega) \mapsto \omega - t$, an open map. If $C_1 = \emptyset$ then $G \equiv 0$ and $g = 0$. A nonempty open subset of $\mathbb{T}$ contains an interval. $\square$

Write $z = e^{2\pi i t}$ and

$$(10) \quad Q_\theta(z) = Cz^2 + Az + Be^{-2\pi i \theta}, \quad P_\theta(t) = z^{-1} Q_\theta(z).$$

Only the phase of the constant term of Q_θ depends on θ ; this drives everything below. Let $N(\theta)$ denote the number of zeros of Q_θ in the open unit disc, counted with multiplicity, defined whenever Q_θ has no zero on the unit circle. Put

$$(11) \quad E_\partial = \{\theta : Q_\theta \text{ has a zero on } |z| = 1\}.$$

3. THE CENTRAL IDENTITY: THE ROOT COUNT

The following proposition is the heart of both routes. We give two independent proofs, with different hypotheses and different costs.

Proposition 7 (Root count). *Let θ be a live fibre with $\theta \notin E_\partial$. Then*

$$N(\theta) = j + 1.$$

This holds (i) unconditionally, if Zg is continuous; and (ii) for arbitrary $g \in L^2(\mathbb{R})$, assuming Theorem 5.

Both proofs pass through the same normalisation. Define

$$(12) \quad m_\theta(t) = e^{2\pi i \theta t} e^{\pi i t(t-1)},$$

so that $|m_\theta| = 1$ and $m_\theta(t+1)/m_\theta(t) = e^{2\pi i(t+\theta)}$. Hence $V_\theta = G_\theta/m_\theta$ is 1-periodic by (6). From $t(t-1) - (t-a)(t-a-1) = 2at - a^2 - a$ we get

$$\frac{m_\theta(t)}{m_\theta(t-a)} = e^{2\pi i \theta a} e^{2\pi i a t} e^{-\pi i(a^2+a)},$$

and the non-periodic exponentials cancel, since $e^{-2\pi i(a+j)t} \cdot e^{2\pi i a t} = e^{-2\pi i j t}$. Therefore (7) becomes

$$(13) \quad V_\theta(t-a) = \Lambda_\theta e^{-2\pi i j t} P_\theta(t) V_\theta(t), \quad \Lambda_\theta = -D^{-1} e^{2\pi i \theta a} e^{-\pi i(a^2+a)},$$

with $\Lambda_\theta \neq 0$ constant. Since the fibre is live, $V_\theta \neq 0$ almost everywhere; put $W_\theta = V_\theta/|V_\theta|$. Because $\theta \notin E_\partial$, P_θ is zero free on $\mathbb{T}$, so (13) gives

$$(14) \quad W_\theta(t-a) = \xi_\theta(t) W_\theta(t), \quad \xi_\theta(t) = \frac{\Lambda_\theta}{|\Lambda_\theta|} e^{-2\pi i j t} \frac{P_\theta(t)}{|P_\theta(t)|}.$$

The map $\xi_\theta : \mathbb{T} \rightarrow \mathbb{T}$ is real analytic, and by the argument principle

$$(15) \quad \deg \xi_\theta = -j + \text{wind}(P_\theta, 0) = -j - 1 + N(\theta),$$

because $P_\theta = z^{-1} Q_\theta$, the factor z^{-1} contributes winding number -1 , and $Q_\theta(e^{2\pi i t})$ has winding number $N(\theta)$. Proposition 7 is thus equivalent to $\deg \xi_\theta = 0$.

Proof of Proposition 7(i): continuous windows. Here G_θ is continuous and, by Lemma 6, nowhere zero on a live fibre; hence V_θ is continuous and nowhere zero and $W_\theta : \mathbb{T} \rightarrow \mathbb{T}$ is a continuous circle map, so it has a degree. Taking degrees in (14) and using that precomposition with a rotation does not change the degree of a circle map,

$$\deg W_\theta = \deg \xi_\theta + \deg W_\theta,$$

whence $\deg \xi_\theta = 0$. Now apply (15). □

Proof of Proposition 7(ii): arbitrary L^2 windows. For merely measurable W_θ no degree is defined and the previous argument is unavailable. Suppose $\deg \xi_\theta \neq 0$ and let $(U_\theta f)(t) = \xi_\theta(t) f(t-a)$. Since ξ_θ is real analytic with

nonzero degree, Theorem 5 applies and U_θ has no eigenvectors. On the other hand, conjugating (14),

$$U_\theta \overline{W_\theta} = \xi_\theta(t) \overline{W_\theta(t-a)} = \overline{W_\theta(t)},$$

so $\overline{W_\theta}$ is a unimodular — in particular nonzero — eigenvector of U_θ with eigenvalue 1. This is a contradiction, so $\deg \xi_\theta = 0$; apply (15). $\square$

Remark 8. The two proofs are genuinely different, and neither subsumes the other. Route (i) is unconditional but needs continuity, which is exactly what makes a degree available; route (ii) needs no regularity but purchases the same conclusion from a spectral theorem. Note the inversion: in route (i) the vanishing $\deg \xi_\theta = 0$ is *forced* by the mere existence of a continuous 1-periodic solution, whereas in route (ii) it is *deduced* from the non-existence of eigenvectors when the degree is nonzero. This is why degree-based obstructions for cocycles over irrational rotations (Anzai [1]; [9]) are useless in the continuous category here and decisive in the measurable one.

4. THE MEAN CONDITION

We record a measurable-coboundary fact, which replaces the appeal to continuity in the evaluation of the fibre mean.

Lemma 9. *Let R be an ergodic measure-preserving transformation of a probability space. Suppose $\varphi \in L^1$ and u is finite and measurable almost everywhere with $\varphi = u \circ R - u$ almost everywhere. Then $\int \varphi = 0$.*

Proof. By Birkhoff's theorem $\frac{1}{n} \sum_{k=0}^{n-1} \varphi \circ R^k \rightarrow \int \varphi$ almost everywhere. The left side equals $(u \circ R^n - u)/n$. For every $\varepsilon > 0$,

$$\mu(|u \circ R^n - u| > n\varepsilon) \leq 2\mu(|u| > n\varepsilon/2) \rightarrow 0,$$

since $u \circ R^n$ and u have the same distribution. So the Birkhoff averages converge to zero in measure, and $\int \varphi = 0$. $\square$

On a live fibre, Lemma 6 lets us take logarithms in (9), the values on a null set being defined arbitrarily. Since $\log |P_\theta| \in L^1(\mathbb{T})$, Lemma 9 yields

$$(16) \quad \int_0^1 \log |P_\theta(t)| dt = \log |D|.$$

(For continuous Zg one may instead integrate $\log |G_\theta|$, which is continuous and 1-periodic, over a period; the G terms cancel by translation invariance of Lebesgue measure alone. Lemma 9 is what makes the step available without continuity.)

Suppose $N(\theta) = 1$ and let $\zeta_{\text{out}}(\theta)$ be the zero of Q_θ outside the closed unit disc. Since $\int_0^1 \log |z^{-1}| = 0$, Jensen's formula together with (16) gives

$$(17) \quad \log |D| = \log |C| + \log |\zeta_{\text{out}}(\theta)|, \quad \text{i.e.} \quad |\zeta_{\text{out}}(\theta)| = \frac{|D|}{|C|}.$$

This is a quantity that does not depend on θ — the fixed point of both endgames.

5. TWO ENDGAMES, AND THE PROOFS

Proposition 7 disposes of $|j| \geq 2$ at once, since $N(\theta) \in \{0, 1, 2\}$. The case $j = 0$ requires (17), and can be closed in two ways.

5.1. Fixed-radius counting. This endgame is measure-theoretic and needs no continuity.

Lemma 10 (Fixed-radius counting). *For each $\rho > 0$ there are at most two values of $\theta \in \mathbb{T}$ for which Q_θ has a zero of modulus ρ .*

Proof. Suppose $Q_\theta(\rho e^{i\phi}) = 0$. Taking absolute values, $\rho |A + C\rho e^{i\phi}| = |B|$, and after squaring

$$\Re(C\bar{A}e^{i\phi}) = \frac{|B|^2/\rho^2 - |A|^2 - |C|^2\rho^2}{2\rho}.$$

Because $AC \neq 0$ by (4), this has at most two solutions ϕ modulo 2π . For each such zero $z = \rho e^{i\phi}$ the equation $e^{-2\pi i\theta} = -(Cz^2 + Az)/B$ determines at most one $\theta \in \mathbb{T}$. $\square$

In particular E_∂ has at most two elements (take $\rho = 1$). Combining (17) with Lemma 10 at $\rho = |D|/|C|$:

$$(18) \quad \{\theta : \theta \text{ live and } N(\theta) = 1\} \quad \text{has at most two elements.}$$

5.2. Vieta rigidity. This endgame is topological and needs an interval of live fibres, which Lemma 6 supplies in the continuous category.

Lemma 11 (Vieta rigidity). *Suppose $j = 0$, Zg is continuous, and $I \subseteq C_1 \setminus E_\partial$ is an interval of positive length. Then no dependence (3) exists.*

Proof. By Proposition 7, $N(\theta) = 1$ on I , so Q_θ has one root $\zeta_{\text{in}}(\theta)$ inside and one root $\zeta_{\text{out}}(\theta)$ outside the unit circle, and both depend continuously on θ because the two roots stay separated. Write $r = |D|/|C|$ and $S = -A/C$. By Vieta,

$$\zeta_{\text{in}} + \zeta_{\text{out}} = S \text{ is constant on } I, \quad \zeta_{\text{in}}\zeta_{\text{out}} = (B/C)e^{-2\pi i\theta}.$$

By (17), $|\zeta_{\text{out}}| \equiv r$, hence $|\zeta_{\text{in}}| = |B|/(|C|r)$ is constant too, and therefore $|\zeta_{\text{out}} - S| = |\zeta_{\text{in}}|$ is a positive constant r' (positive because $\zeta_{\text{in}}\zeta_{\text{out}} \neq 0$, as $B \neq 0$). Thus $\zeta_{\text{out}}(\theta)$ lies in

$$\{|z| = r\} \cap \{|z - S| = r'\},$$

the intersection of two circles with distinct centres, since $S \neq 0$ as $A \neq 0$. That set has at most two points. A continuous function on the interval I with values in a finite set is constant, so ζ_{out} , and hence ζ_{in} , is constant on I ; therefore the product $(B/C)e^{-2\pi i\theta}$ is constant on I , which is false since $B \neq 0$ and I has positive length. $\square$

Proof of Theorem 1. Suppose a nontrivial dependence (3) exists; by (4) all coefficients are nonzero. By Lemma 6, C_1 is nonempty and open, hence contains an interval; removing the at most two points of E_∂ leaves an interval I of positive length.

If $|j| \geq 2$, Proposition 7(i) gives $N(\theta) = j + 1 \notin \{0, 1, 2\}$ for $\theta \in I$, which is absurd. If $j = 0$, Lemma 11 applies to I and gives a contradiction directly. Since $j \notin \{-1, 1\}$ by hypothesis, these cases are exhaustive.

For the final assertion, a nonzero Schwartz g — more generally a nonzero $g \in W(C, \ell^1)$ — has $\sum_n |g(t - n)|$ locally uniformly convergent with continuous terms, so Zg is continuous. The case $a = \sqrt{2}$, $j = 0$ is Λ_0 after the shift $(\sqrt{2}, \sqrt{2})$, which is [8, Conjecture 2]. $\square$

Proof of Theorem 2. Suppose a nontrivial dependence (3) exists; again $ABCD \neq 0$. By Lemma 10 the set E_∂ is finite, so almost every live fibre lies outside it, and by (8) the live fibres have positive measure.

If $|j| \geq 2$: since $N(\theta) \in \{0, 1, 2\}$ we have $N(\theta) \neq j + 1$, so Proposition 7(ii) shows every live fibre lies in E_∂ up to a null set. As E_∂ is finite this contradicts positivity of the measure of live fibres.

If $j = 0$: Proposition 7(ii) excludes every live fibre outside E_∂ with $N(\theta) = 0$ or $N(\theta) = 2$, and (18) says the live fibres with $N(\theta) = 1$ number at most two. So the live-fibre set is finite up to a null set, again contradicting (8).

Since $j \notin \{-1, 1\}$, these cases are exhaustive, and no dependence exists. $\square$

6. WHY $j = \pm 1$ SATURATES BOTH ROUTES

For $j = \pm 1$ Proposition 7 gives $N(\theta) \in \{0, 2\}$: both roots of Q_θ lie on the same side of the unit circle. Jensen's formula then returns $\log |C|$ (if $N = 2$) or $\log |B|$ (if $N = 0$), so (17) degenerates to the single θ -independent condition $|C| = |D|$, respectively $|B| = |D|$. Both endgames lose their input: the radius $\rho = |D|/|C|$ of Lemma 10 is no longer pinned to a root, and the Vieta rigidity of Lemma 11 has nothing to act on.

Two further routes are unavailable. First, the degree-based obstructions for cocycles over irrational rotations (Anzai [1]; [9]) require a cocycle of nonzero degree, whereas in the continuous category (15) pins $\deg \xi_\theta = 0$ for *every* j : the degree is forced by the mere existence of a continuous 1-periodic solution. (For merely measurable windows there is no degree, nothing forces 0, and these theorems do apply — this is exactly the asymmetry exploited in route (ii) and recorded in Remark 8.)

Second, no single fibre obstructs. For $N = 2$ one has $\Lambda_\theta e^{-2\pi i j t} P_\theta = \Lambda_\theta C(1 - \zeta_1/z)(1 - \zeta_2/z)$ up to a character, so log of the symbol has Fourier support in nonpositive frequencies with geometrically decaying coefficients, and the cohomological equations $\hat{v}(n)(e^{2\pi i n a} - 1) = \hat{\psi}(n)$ are solvable for all $n \neq 0$, the mean condition $\hat{\psi}(0) = 0$ being exactly (16). For badly approximable a the small divisors are dominated by the geometric decay.

Root location is likewise unobstructed: with $\zeta_1 + \zeta_2 = S$ fixed and $|\zeta_1 \zeta_2|$ constant, the Schur–Cohn conditions are satisfiable on a short interval.

Consequently a proof for $j = \pm 1$ must use global input that neither route needs: the quasi-periodicity (1) makes G a section of a line bundle over $\mathbb{T}^2$ of Chern number one, so every continuous Zak transform vanishes somewhere and dead fibres necessarily exist; and the fibrewise solutions must be glued into a single continuous G with matching behaviour on ∂C_1 .

7. RELATION TO THE WORK OF GUAN AND OKOUDJOU

While this work was being completed, Guan and Okoudjou [6] posted a proof of the HRT conjecture for the symmetric $(2n+1, 2)$ configurations $\Lambda_n = \{(0, k) : -n \leq k \leq n\} \cup \{(a, b), (a, -b)\}$ with arbitrary L^2 windows and $ab \neq 0$ (their Theorem 1; more generally whenever the collinear spacings are commensurable). Their Corollary 1 reads: *let $0 \neq g \in L^2(\mathbb{R})$ be real-valued and let $\tilde{\Lambda} = \{(0, 0), (0, 1), (s, 0), (a, b)\}$ consist of four distinct points with $a, b, s \neq 0$; then the HRT conjecture holds for $\tilde{\Lambda}$ and g .* For a real-valued window, conjugating a dependence sends $b \mapsto -b$ and produces the symmetric companion point, placing the configuration inside a symmetric $(3, 2)$ configuration covered by their Theorem 2. Their Corollary 2 then extends this to $h = e^{2\pi i \varphi} g$ for g real-valued and φ a real polynomial of degree at most 2, the quadratic part being absorbed by a metaplectic operator.

Note that Λ_0 is an instance of $\tilde{\Lambda}$ (take $s = 1$, $(a, b) = (\sqrt{2}, \sqrt{2})$), so their results apply to it directly. The relationship to the present paper should be stated plainly, in both directions.

- **Their result is far more general in the configuration.** It covers every four-point set, whereas Theorems 1 and 2 cover only the resonant family $\Lambda_{a,j}$. It is also unconditional, and its proof is dramatically shorter than the development above. *For windows of the form $e^{2\pi i \varphi} g$ with g real-valued and $\deg \varphi \leq 2$, the Λ_0 case is settled by [6]* — a substantially larger class than “real-valued” alone. We claim no priority for it.
- **Our results are more general in the window class, and that is where the open problem lies.** The conjugation requires $\bar{g} = g$, and the metaplectic absorption requires φ to be a polynomial of degree at most 2. Guan and Okoudjou identify the obstruction themselves: writing $g = |g|e^{2\pi i \varphi}$ with φ merely measurable, their Remark 2 observes that Corollary 1 settles $|g|$ and Corollary 2 settles quadratic φ , so that *“the essential remaining problem is to understand the effect of more general phase functions.”* Theorems 1 and 2 impose *no condition whatsoever on the phase*: a nonzero Schwartz window with, say, a cubic phase $e^{2\pi i x^3} e^{-x^2}$ is covered by Theorem 1 and by neither corollary of [6]. Neither of our theorems is implied by [6].

- **Consequently the Heil–Speegle subconjecture is not settled** by [6]. [8, Conjecture 2] is stated for $g \in \mathcal{S}(\mathbb{R}) \setminus \{0\}$, and a Schwartz function may carry an arbitrary smooth phase. Theorem 1 settles it as stated.

Neither result dominates: [6] is stronger in configuration generality and unconditionality, this paper is stronger in window class for one resonant family.

We record what seems to us the most promising synthesis, since the two developments meet more closely than the surface suggests. Remark 2 of [6] asks for control of the *phase* of the window; the Zak fibration used here is a coordinate system in which the phase separates cleanly from the modulus. Indeed (9) is a cocycle in $|G_\theta|$ alone — the $|g|$ half, which Jensen’s formula (17) resolves — while (14) is a cocycle in $W_\theta = V_\theta/|V_\theta|$ alone — the $e^{2\pi i \varphi}$ half. The quadratic phase absorbed metaplectically in their Corollary 2 is, on this side, exactly the normalising factor m_θ of (12). Moreover their mechanism and ours act on the same object: their Theorem 1 turns on the collinear trigonometric polynomial factoring into linear terms in $\omega = e^{2\pi i a t}$, and our Q_θ is precisely such a factorisation, with $N(\theta)$ counting the factors inside the disc.

This suggests replacing the one quoted input of Theorem 2. Our appeal to Theorem 5 is needed only because a measurable W_θ has no degree; the product estimates of Demeter–Zaharescu used in [6], which bound $\prod_n |P(y+n)|$ above and below on a set of measure $\geq 1 - \varepsilon$, never require one. Iterating (9) gives $|G_\theta(t - Na)| = |D|^{-N} \prod_{n < N} |P_\theta(t - na)| |G_\theta(t)|$, so a two-sided bound on that product on a set of positive measure would contradict $G_\theta \in L^2$ directly. Our (16) is the logarithmic — hence weaker — form of the same statement: it pins the geometric mean of the product, where the estimates of [6] pin the partial products themselves. Establishing the two-sided bound uniformly in the fibre parameter θ is the technical crux, and is the direction we regard as most likely to make Theorem 2 unconditional.

8. FORMAL VERIFICATION

Route (ii), the L^2 route, has been formalised in Lean 4 against Mathlib and is machine checked. We describe precisely what is and is not covered, since a formalisation claim is worthless without its boundary.

The development comprises 25 modules, about 19,300 lines and 1,065 declarations, of which 330 are in the capstone module; there is no `sorry` anywhere in it. Every theorem cited below has axiom footprint exactly `[propext, Classical.choice, Quot.sound]`, i.e. Lean’s standard classical foundation and nothing else. The following are proved outright:

- the Zak reduction of §2, the covariance (2) and the modulus cocycle (9), for arbitrary real shifts and arbitrary j , with no compact-support or decay hypothesis;

- the Birkhoff coboundary lemma (Lemma 9) and the mean condition (16);
- the argument principle in the form actually required, (15): for the quadratic Q_θ with no root on the unit circle, an explicit lift is constructed and its winding number computed, giving $\deg \xi_\theta = -(j + 1) + N(\theta)$ (`exists_lift_quadratic`). Mathlib has no winding number, no argument principle and no $\pi_1(S^1) \cong \mathbb{Z}$; none is needed, because only this one family of loops occurs;
- the fixed-radius counting lemma (Lemma 10), Jensen's consequence (17), and the endgame of Theorem 2 from Proposition 7(ii);
- the three-point input of §2, for this configuration (`hthree_lambdaZero`): every three-point subset of Λ_0 is handled outright, so the reduction to $ABCD \neq 0$ is *proved* rather than cited, and the final theorem carries no three-point hypothesis;
- an unconditional consequence of a hypothetical dependence, `lambdaZero_dependence_forces_n` which requires no external input at all.

Theorem 5 itself is *not* proved. Its conclusion — Lebesgue maximal spectral type — is not currently expressible in Mathlib, which has no maximal spectral type, no spectral measure for unitary operators and no notion of Lebesgue spectrum. It is therefore carried as an explicit Lean hypothesis `ILRStatement`, a `def` of type `Prop` discharged by the caller, *not* as an `axiom`: consumers such as `heil_speegle_lambda_zero_L2_of_ILR` and `lambda0Translates_linearIndependent_of_ILR` take it as an argument, so no `sorryAx` and no additional axiom enters any footprint. The statement is pinned to the loops that actually arise (`quadLoop`), so that it is not accidentally stronger than [9]. One nontrivial fragment of it *is* proved outright: the character case $\xi(t) = e^{2\pi i d t}$, $d \neq 0$ (`ILR_character`), via Parseval and the invariance of $|\widehat{W}|$ along arithmetic progressions of step d .

Route (i) of Proposition 7 — the unconditional continuous-window argument, and hence Theorem 1 — is **not** formalised. The winding-number infrastructure it would need is present (`windOf`, `IsLoopLift`, `windOf_add`, `exists_lift_char`), but the invariance of the degree under precomposition with a rotation, and the continuous fibre dichotomy, are not. Formalising route (i) would remove the one quoted input from the formal development and is the natural next step.

A separate machine check certifies statement fidelity: the theorem `lambda0_independent_of_reducti` is verified with `Comparator` [3], which confirms that the claimed proof establishes exactly the statement in the trusted statement file, using no axioms beyond the three permitted ones. `Comparator` certifies fidelity and axiom footprint; it cannot and does not upgrade a conditional theorem to an unconditional one.

9. ATTRIBUTION

This paper builds on the program of Oussa [14], and we want the division of credit to be unambiguous. The following are already present there, for this configuration family and for Schwartz windows: the Zak-transform reduction (5) (his Lemma 2.1); zero propagation (his Lemma 2.2 and Theorem 3.3); the non-ergodicity of the relevant translation, the resulting fibration of $\mathbb{T}^2$ into invariant circles, and the orbit-saturation dichotomy underlying Lemma 6 (his Proposition 3.1); the vanishing of the averaged logarithmic growth (his Proposition 3.2); and the integrality and constancy of the winding number along lattice loops (his Lemma 6.1). Specifically, the proof of his Proposition 6.2 derives an identity which, specialised to $d = 1$, fourth point $(a, a + j)$, $a = (-1, 1)$ and $m = 1$, yields the winding value; the winding lemma there is in turn credited to suggestions of C. Heil, K. Okoudjou and R. Jadhav. Oussa works throughout with Schwartz windows and derives *necessary* dynamical, cohomological and arithmetic conditions in the infinite proper orbit case, rather than claiming a proof of that case; he names Λ_0 as the unresolved resonant case.

Accordingly we claim only the following.

- (i) The derivation of $\deg \xi_\theta = 0$ using *continuity alone*, with no smoothness hypothesis, via the invariance of the degree of a circle map under precomposition with a rotation (Proposition 7(i)).
- (ii) The reading of that identity as a *root count* $N(\theta) = j + 1$, which immediately settles $|j| \geq 2$ by a range count on N . No root count of a quadratic in the disc appears in [14].
- (iii) The evaluation of the fibre mean by Jensen's formula so as to produce a quantity constant across fibres, (17). Jensen's formula does not appear in [14].
- (iv) The two endgames: the fixed-radius counting lemma (Lemma 10), which is what settles Λ_0 in L^2 , and the Vieta rigidity (Lemma 11), which settles it in the continuous category. Vieta's relations do not appear in [14].
- (v) The removal of the regularity hypothesis altogether via Theorem 5 (Proposition 7(ii)), and the observation of Remark 8 that the continuous and measurable categories require opposite arguments for the same identity.
- (vi) The formal verification described in Section 8.

Items (iii), (iv) and (v) are, to our knowledge, new. For the real-valued case of Λ_0 see the discussion of [6] in Section 7; priority there is theirs.

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